Unified Prompt-Profile Report: Setup, Views, Results, and Mechanism

2026-04-09 22:55 UTC

Executive Summary

- This is the single current prompt-profile report surface.
  - Regression stays on the natural prompt-disjoint split with natural sampling.
  - Classification stays on balanced train with natural test.
  - The same note now carries setup, predictor views, metrics, results, and the plain-English metadata explanation in one place.
- Regression read:
  - `mean_relative_length` is useful as a screening score, not as a clean calibrated regressor.
  - On `top_20p_capture`, ensemble beats prompt length on `GPQA`, `MMLU-Pro`, and `LiveCodeBench`.
- Classification read:
  - `majority_s_0.5` is still the cleaner deployment-facing head.
  - The best single global activation surface is still `ensemble h256 d1`.
- Metadata read:
  - The old reported metadata baseline is only prompt length on this fixed-budget run.
  - Prompt-only cues are strong mostly because prompt surface structure already carries much of the same long-completion signal.
  - `GPQA` is the main exception where raw length is weak and prompt structure matters more.

Setup

Common rollout contract

- model: `Qwen/Qwen3-1.7B`
- decode policy: `temperature=0.2`, `num_generations=4`, `max_tokens=30000`

- features: prompt-prefill last-token activations over all saved layers (`28 x 2048`)
- frozen checkpoint rule for headline tables: `best_loss`
- compute contract for the latest mechanism audit: GPU node, `2` GPUs

Dataset surface

Dataset	Regression train prompts	Regression test prompts
`GPQA`	158	40
`AIME`	48	12
`MATH-500`	400	100
`MMLU-Pro`	640	160
`LiveCodeBench`	640	160

Split contract

- Regression: natural prompt-disjoint train/test split with natural sampling.
- Classification: train balanced by downsampling, test kept at natural prevalence.
- This split is now part of the settled object, not an open choice:
  - regression should not be balanced because the target is continuous;
  - classification stays balanced at train time because the target is binary and rare on several datasets.

Predictor views

View	Definition
`prompt length`	train-fit linear scorer on `prompt_token_count`; on this run `effective_max_tokens` is fixed so the metadata baseline reduces to prompt length
`prompt-shape linear`	linear model on `prompt_token_count`, `log_token_length`, `char_length`, `newline_count`, `digit_count`, `dollar_count`, `choice_count`
`prompt-shape tree`	nonlinear tree on the same seven prompt-surface features
`last_layer`	one MLP on the final prompt-prefill layer; regression keeps the locked `h128 d1`, binary control uses `h256 d2`
`ensemble`	one MLP per saved layer over the full `28 x 2048` prompt-prefill surface; regression keeps locked `h128 d1` and aggregates by `mean_prob`; classification recommendation is `h256 d1` and aggregates by `vote_fraction`

Hyperparameters

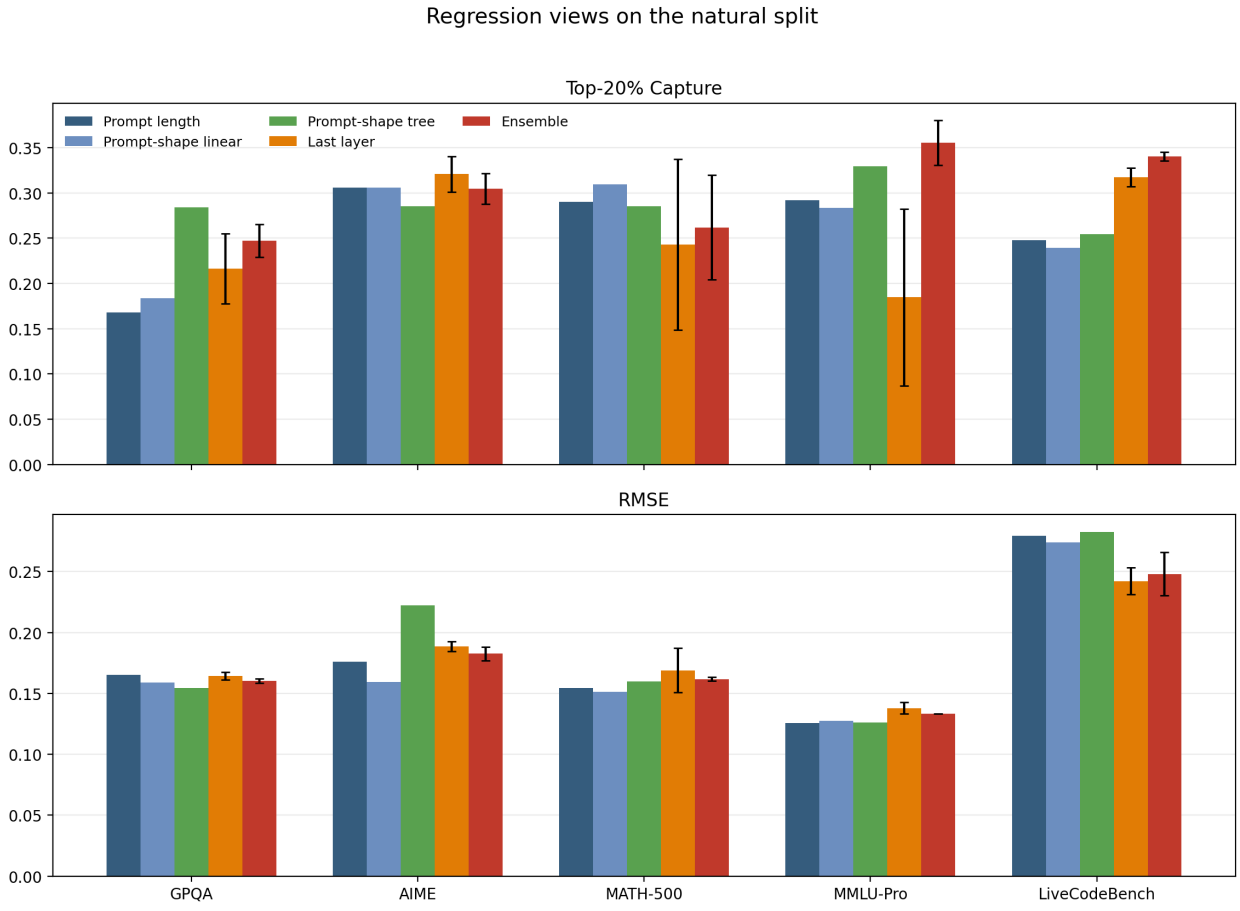
- Regression activation probes keep the locked full-train object: `h128 d1`, `dropout=0.1`, seeds `0,1,2`, `epochs=10`, `batch_size=256`, `lr=1e-4`, `weight_decay=0.1`.
- Classification capacity sweep compared `h128 d1`, `h256 d1`, and `h256 d2` with the same optimizer surface: `epochs=15`, `batch_size=256`, `lr=1e-4`, `weight_decay=0.05`, `dropout=0.1`.

- The current recommended binary activation surface is `ensemble h256 d1`. The current last-layer control is `h256 d2`.

Metrics In Plain English

- `top_20p_capture`: sort held-out prompts by score, keep the top `20%`, and ask how much of the actual long-completion mass those prompts contain.
- `RMSE`: ordinary pointwise prediction error on aligned prompt pairs. Lower is better.
- `PR-AUC`: ranking quality for the positive class. This is the main binary metric because several datasets are highly imbalanced.
- Positive precision / recall / `F1`: threshold metrics for how the chosen operating point behaves on the positive class.
- Macro `F1`: average `F1` across both classes, so it penalizes collapsing to one class.
- `Spearman`: monotone-ordering diagnostic only. It is not the task definition.

Regression Results

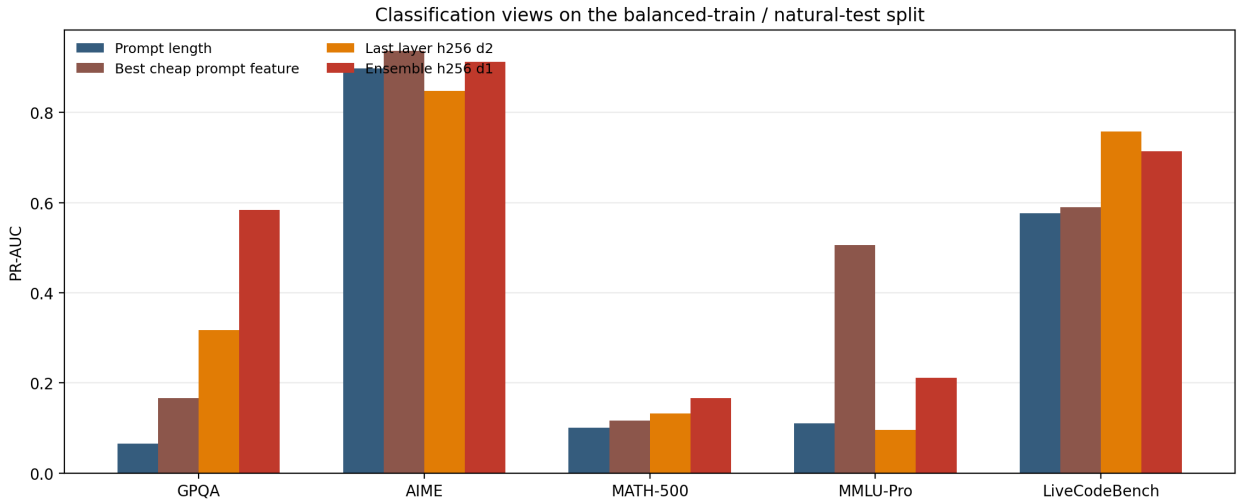


Dataset	Prompt top_20	Last-layer top_20	Ensemble top_20	Prompt RMSE	Last-layer RMSE	Ensemble RMSE
GPQA	0.168	0.216 +/- 0.039	0.247 +/- 0.018	0.165	0.164 +/- 0.003	0.160 +/- 0.002
AIME	0.306	0.321 +/- 0.020	0.305 +/- 0.017	0.176	0.189 +/- 0.004	0.183 +/- 0.006
MATH-500	0.290	0.243 +/- 0.094	0.262 +/- 0.058	0.154	0.169 +/- 0.018	0.162 +/- 0.002
MMLU-Pro	0.292	0.185 +/- 0.098	0.355 +/- 0.025	0.126	0.138 +/- 0.005	0.133 +/- 0.000
LiveCodeBench	0.248	0.317 +/- 0.010	0.340 +/- 0.005	0.279	0.242 +/- 0.011	0.248 +/- 0.018

Plain read:

- Screening is the right primary read for this lane.
- Ensemble is strongest on GPQA, MMLU-Pro, and LiveCodeBench.
- Prompt length still wins on calibration for AIME, MATH-500, and MMLU-Pro.
- So mean_relative_length is useful as a long-completion screening score, but still mixed as a calibrated regression target.

Classification Results



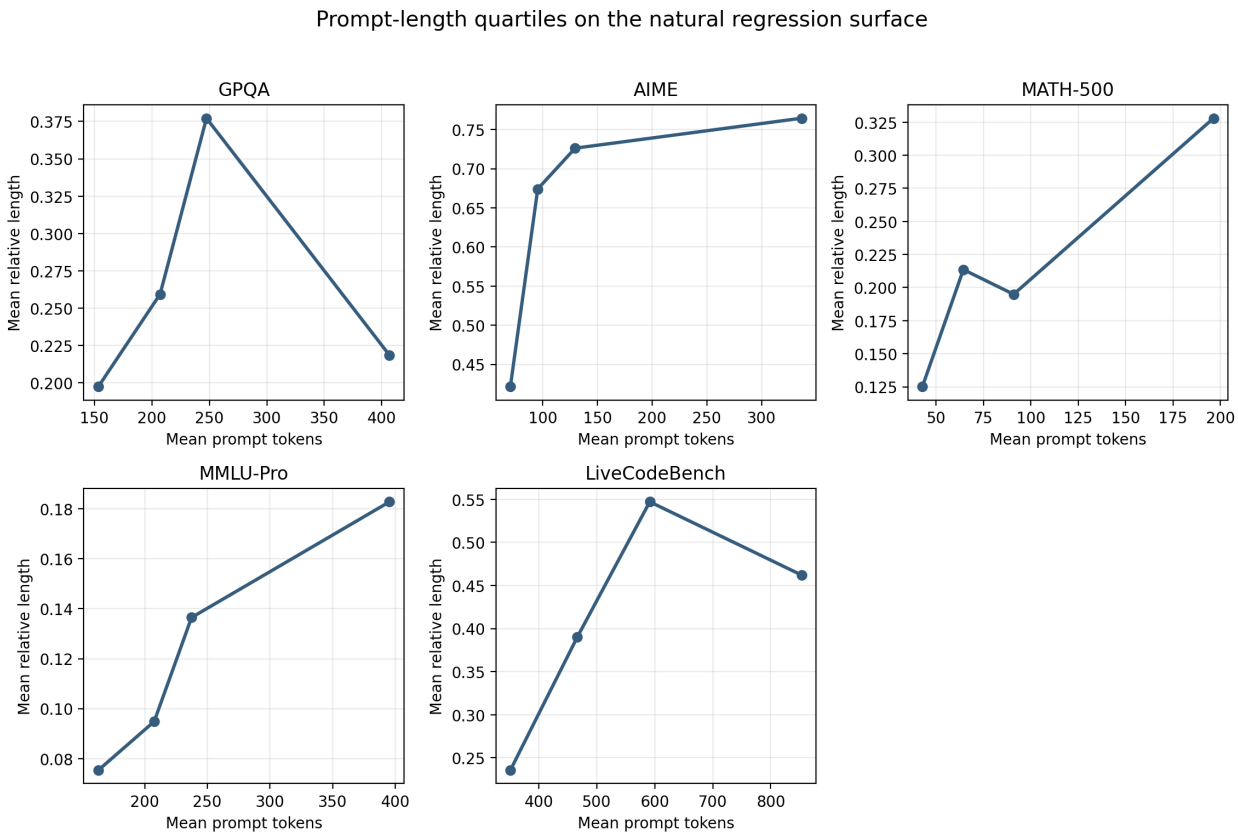
Dataset	Prev.	Prompt PR-AUC	Best prompt-only	Ensemble PR-AUC	Prec.	Rec.	Pos. F1	Macro F1
GPQA	0.0500	0.066	0.167	0.583	0.214	1.000	0.345	0.607
AIME	0.5833	0.898	0.937	0.912	0.905	0.476	0.564	0.614
MATH-500	0.0400	0.100	0.117	0.166	0.054	0.667	0.100	0.439
MMLU-Pro	0.0125	0.110	0.506	0.212	0.125	1.000	0.218	0.575
LiveCodeBench	0.3375	0.576	0.590	0.714	0.614	0.654	0.514	0.581

Plain read:

- majority_s_0.5 remains the cleaner finished head.
- ensemble h256 d1 is still the best single global activation surface by PR-AUC.

- Threshold metrics stay recall-heavy on rare-positive datasets because train balance and test prevalence differ.
- That is why `PR-AUC` remains the headline metric and precision / recall / `F1` stay diagnostic.

Why The Metadata Baseline Is Strong



Prompt-only cues by dataset

Dataset	Best regression prompt-only cue	Best classification prompt-only cue
`GPQA`	Newline count 0.242	Newline count 0.167
`AIME`	Character length 0.332	Dollar count 0.937
`MATH-500`	Prompt length 0.290	Dollar count 0.117
`MMLU-Pro`	Dollar count 0.300	Newline count 0.506
`LiveCodeBench`	Prompt length 0.248	Character length 0.590

Plain-English interpretation

- The old "metadata baseline" is narrow: on this run it is really just prompt length, because `effective_max_tokens` is fixed at `30000`.

- Prompt length predicts completion length mostly because, on these datasets, longer prompts usually mean the prompt itself contains more work.
- That story is strongest on `AIME`, `MATH-500`, and much of `LiveCodeBench`.
- `MMLU-Pro` has the same effect, but more weakly.
- `GPQA` is the honest exception: the longest prompts are not the riskiest prompts there, which is why prompt-shape models help more than raw length.
- Classification inherits some of the same prompt-only signal because `majority_s_0.5` is still largely asking whether rollouts cross about half the fixed budget.
- That is why prompt-only features can be strong on both heads without implying a hidden implementation bug.

Quartile read

- `GPQA`: 153 -¿ 0.197, 207 -¿ 0.259, 248 -¿ 0.377, 407 -¿ 0.218
- `AIME`: 70 -¿ 0.422, 96 -¿ 0.674, 130 -¿ 0.726, 336 -¿ 0.765
- `MATH-500`: 43 -¿ 0.125, 64 -¿ 0.214, 91 -¿ 0.195, 196 -¿ 0.328
- `MMLU-Pro`: 163 -¿ 0.075, 208 -¿ 0.095, 237 -¿ 0.137, 395 -¿ 0.183
- `LiveCodeBench`: 351 -¿ 0.236, 467 -¿ 0.390, 591 -¿ 0.547, 853 -¿ 0.462

Bottom Line

- Regression answer: keep the natural split / natural sampler object and read it screening-first.
- Classification answer: keep balanced train / natural test and use `ensemble h256 d1` as the current default activation surface.
- Metadata answer: prompt-only predictors are strong mainly because prompt surface structure already acts as a workload proxy on these fixed-budget targets.
- Activation answer: the current surface shows lift over raw prompt length on some slices, but it still does not prove general lift over strong prompt-only controls.
- Next honest check: a stronger prompt-shape baseline plus residualized activation lift on the natural regression head.

Artifacts

- unified bundle: `outputs/prompt_profile_unified_report_20260409/`
- source notes behind this synthesis:

- docs/prompt-profile-combined-audit-2026-04-05.md
- docs/prompt-profile-binary-capacity-controls-2026-04-04.md
- docs/prompt-profile-natural-regression-rerun-2026-04-05.md
- docs/prompt-profile-length-mechanism-2026-04-09.md